**LIST O PUBLICATIONS**

Surajit Das

## I. Published and Accepted Peer-Reviewed Publications

**1. A resource-efficient deep learning pipeline for brightfield live-cell segmentation under constrained computational environments** (2026). Published in Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization (**Q2**). | Journal article | DOI: https://doi.org/10.1080/21681163.2026.2723013

**2. Microplastic Contamination and Detection in Food Systems: A Review of Machine Learning, Traditional Methods, & Other Relevant Factors** (2025). Published in Microchemical Journal (**Q1**). | DOI | DOI: https://doi.org/10.1016/j.microc.2025.115440

**3. Intelligent Software System for Low-Cost, Brightfield Segmentation: Algorithmic Implementation for Cytometric Auto-Analysis** (2025). Presented at IEEE DESE 2025; published in IEEE Xplore. | Conference | IEEE Xplore | Certificate | DOI: 10.1109/DeSE68208.2025.11367949

**4. Educators on the Frontline: Philosophical and Realistic Perspectives on Integrating ChatGPT into the Learning Space** (2025). Accepted; in production at Voprosy Obrazovaniya (**Q2**). | Journal | arXiv link supplied | DOI: Not available till now

**5. Predicting ChatGPT Use in Assignments: Implications for AI-Aware Assessment Design** (2025). Presented at T4E 2025, IIT Chennai; Published recently in Springer LNET |
( https://link.springer.com/book/9789819274031#accessibility-information ). | arXiv link supplied | Certificate | DOI not available till now

**6. Advanced Optimization Techniques & Its Application in AI-Powered Breast Cancer Classification** (2023). Published in IEEE Xplore. | IEEE Xplore | DOI: 10.1109/DeSE58274.2023.10099678

## II. Preprints (Most recent First)

**1. Proof-Grounded Patient-Specific Clinical Explanations from Knowledge-Graph Reasoning** (5 October 2026). arXiv preprint, arXiv:2610.06549. DOI: 10.48550/arXiv.2610.06549 | arXiv | DOI: https://doi.org/10.48550/arXiv.2610.06549

**2. Training-Free Clinical Reasoning through Medical Ontologies and Cognitive Mapping: A Symbolic-Probabilistic Knowledge Graph Framework** (28 September 2026). arXiv preprint, arXiv:2609.35298. DOI: 10.48550/arXiv.2609.35298 | arXiv | DOI: https://doi.org/10.48550/arXiv.2609.35298

**3. Disentangling Lung-Cancer CT/LDCT AI: A Systematic Evidence Map of Clinical Tasks, Evidence Chains, and Translational Gaps** (28 September 2026). arXiv preprint, arXiv:2609.35240. DOI: 10.48550/arXiv.2609.35240 | arXiv | DOI: https://doi.org/10.48550/arXiv.2609.35240

**4. From Signals to Causes: A Causal Signal Processing Framework for Robust and Interpretable Clinical Risk Prediction** (27 February 2026). arXiv preprint, arXiv:2602.23977. DOI: 10.48550/arXiv.2602.23977 | arXiv | DOI: https://doi.org/10.48550/arXiv.2602.23977

**5. MAB Optimizer for Estimating Math Question Difficulty via Inverse CV without NLP** (August 2025). arXiv preprint, arXiv:2508.19014. DOI: 10.48550/arXiv.2508.19014 | arXiv | DOI: https://doi.org/10.48550/arXiv.2508.19014

**6. GUI Based Fuzzy Logic and Spatial Statistics for Unsupervised Microscopy Segmentation** (August 2025). arXiv preprint, arXiv:2508.15979. DOI: 10.48550/arXiv.2508.15979 | arXiv | DOI: https://doi.org/10.48550/arXiv.2508.15979